What is the difference between "Nullspace" and "solution space"?

Answer:

In our course MA1513, the term "Nullspace" is used in association to a matrix **M**, and the term "Solution space" is used in association to a homogeneous system **Ax = 0**.

If the matrix **M** is of size m x n, then the nullspace of **M** is a subspace of $\mathbf{R}^n$ that contains all vectors **v** such that, when **v** is pre-multiplied by **M**, the product is the zero vector **0**:

Mv = 0 ---(1)

If the homogeneous system **Ax = 0** has n variables, then the solution space of the system is a subspace of $\mathbf{R}^n$ that contains all vectors **v** that satisfy the system:

Av = 0 ---(2)

We see that (1) and (2) are very similar. In fact, if the coefficient matrix **A** of the system is given by **M**, then the nullspace of **M** is exactly the same set as the solution space of **Ax = 0**.

In practice, when we are given a matrix **M** and asked to find the nullspace, its basis and dimension, we should set up the homogeneous system **Mx = 0** and apply GE/GJE to find the solution space.

To summarise,

- for nullspace, the starting point is a matrix **M**. We are looking for those vectors that are being "transformed" to zero vector by pre-multiplying them with **M**.
- for solution space, the starting point is a homogeneous system **Ax = 0**. We are looking for those vectors that satisfy the system when they are substituted into **Ax = 0**.

Scenario 1

Finding the nullspace and its dimension for the matrix $\mathbf{A} = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ -1 & 2 & -3 \end{bmatrix}$

The approach is to solve the homogeneous equation **Ax = 0**: $\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ -1 & 2 & -3 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

The augmented matrix is: $\begin{bmatrix} 1 & -2 & 3 & 0 \\ 2 & -4 & 6 & 0 \\ -1 & 2 & -3 & 0 \end{bmatrix}$

Reduce the Matrix to Reduced Row Echelon Form (RREF): $\begin{bmatrix} 1 & -2 & 3 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

The first column is a pivot column, while the second and third columns are non-pivot and their corresponding variables, y and z, are **free parameters**.

The number of parameters directly gives us the dimension of the nullspace which is 2 in this example.

From the RREF, we can write the single equation: $x - 2y + 3z = 0$

Let y = s and z = t, where s and t are any real numbers (parameters).

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 2s - 3t \\ s \\ t \end{bmatrix} = s \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} + t \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}$$

Conclusion

1. **The Nullspace:** The nullspace of **A** is the set of all linear combinations of the two vectors, which is the linear span: $\text{span}\left(\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}\right)$.
2. **The Dimension (Nullity):** The basis for the nullspace is $\left\{\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -3 \\ 0 \\ 1 \end{bmatrix}\right\}$. The dimension of the nullspace is the number of vectors in its basis. Our basis consists of two linearly independent vectors. Therefore, the **nullity of A is 2**.

Scenario 2

Finding the nullspace and its dimension for the matrix $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$.

We are solving the system: $\begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

The augmented matrix for this system is: $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 1 & 0 \end{bmatrix}$ which reduce to RREF $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$.

Since the first two columns of the RREF are pivot columns, there is no parameter. This is the key indicator that the nullity will be zero.

Translating the RREF back into equations gives us:

$$1x + 0y = 0 \implies x = 0$$

$$0x + 1y = 0 \implies y = 0$$

The only solution to the system is the trivial solution $\begin{bmatrix} 0 \\ 0 \end{bmatrix}$.

Conclusion

1. **The Nullspace:** The nullspace of **A** contains only one vector, the zero vector in $\mathbb{R}^2$. i.e. nullspace of **A** $= \left\{\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}$. (This is known as the zero space.)

The Dimension (Nullity): The dimension of a vector space is the number of vectors in its basis. The basis for the zero space $\left\{\begin{bmatrix} 0 \\ 0 \end{bmatrix}\right\}$ is the empty set, and its dimension is 0. Therefore, the **nullity of A is 0**.